

2025-11-27 - Valentini (14)

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1 dimensional vlasov

$$\frac{\partial f(x, v, t)}{\partial t} + v \frac{\partial f(x, v, t)}{\partial x} - \frac{eE}{m_e} \frac{\partial f(x, v, t)}{\partial v} = 0$$

Poisson

$$\frac{\partial E(x, t)}{\partial x} = 4\pi e \left[n_0 - \int_{-\infty}^{\infty} f(x, v, t) dv \right]$$

linearize vlasov

$$\frac{\partial f_1}{\partial t} + v \frac{\partial f_1}{\partial x} - \frac{eE_1}{m_e} \frac{\partial f_0}{\partial v} = 0$$

and (considering $|\omega_I/\omega_R| \ll 1$)

$$E_1(x, t) = E_0 e^{\omega_I t} \sin(kx - \omega_R t) \approx E_0 \sin(kx - \omega_R t)$$

$$f_1(x, v, t) = f_0(v) \beta_k \cos(kx - kv t) + \frac{eE_0}{m_e} \frac{\partial f_0}{\partial v} \left[\frac{\cos(kx - \omega_R t) - \cos(kx - kv t)}{\omega_R - kv} \right]$$

with

$$\begin{cases} \beta_k = -\frac{t_0 k}{4\pi e n_0} \\ f_0 = \frac{m_e^{1/2} n_0}{(2\pi k_B T_0)^{1/2}} \exp\left(-\frac{mv^2}{2k_B T_0}\right) \end{cases}$$

At time τ we have

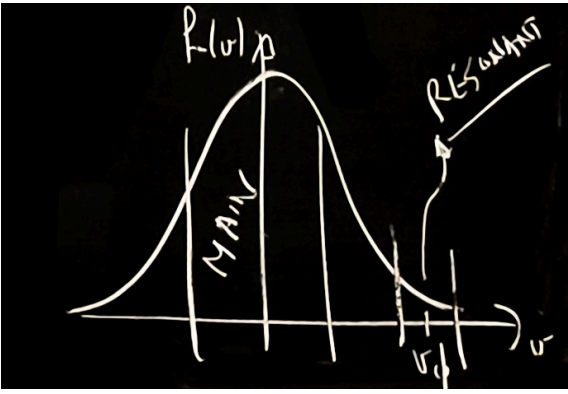
$$\left| \frac{\partial f_0}{\partial v} \right| \approx \left| \frac{\partial f_1}{\partial v} \right|$$

we want to compare them

If they are comparable, then linear approximation fails.

The main effects happens in the resonant region

PIC resonant



We Taylor expand the expression of f_1 in $v = \omega_R/k$.

In particular we take care of the square brackets

$$\begin{aligned} \cos(kx - \omega_R t) - \cos(kx - kv t) &\approx \cos(kx - \omega_R t) - \left[\cos\left(kx - k \frac{\omega_R}{k} t\right) + \right. \\ &\quad \left. + kt \sin\left(kx - k \frac{\omega_R}{k} t\right) \left(v - \frac{\omega_R}{k}\right) + \frac{1}{2} k^2 t^2 \cos\left(kx - k \frac{\omega_R}{k} t\right) \left(v - \frac{\omega_R}{k}\right)^2 + \dots \right] \approx \\ &\approx -kt \sin(kx - \omega_R t) \left(v - \frac{\omega_R}{k}\right) - \frac{1}{2} k^2 t^2 \cos(kx - \omega_R t) \left(v - \frac{\omega_R}{k}\right)^2 \end{aligned}$$

Now we substitute

$$\begin{aligned} f_1(x, v, t) &= f_0(v) \beta_k \cos(kx - kv t) + \frac{eE_0}{m_e} \frac{\partial f_0}{\partial v} \cdot \\ &\quad \cdot \left[\frac{-\cancel{k} t \sin(kx - \omega_R t) \cancel{\left(v - \frac{\omega_R}{k}\right)} - \frac{1}{2} \cancel{k^2} t^2 \cos(kx - \omega_R t) \cancel{\left(v - \frac{\omega_R}{k}\right)^2}}{-\cancel{\left(v - \frac{\omega_R}{k}\right)}} \right] = \\ &= f_0(v) \beta_k \cos(kx - kv t) + \frac{eE_0}{m_e} \frac{\partial f_0}{\partial v} \cdot \\ &\quad \cdot \left[t \sin(kx - \omega_R t) + \frac{1}{2} k t^2 \cos(kx - \omega_R t) \left(v - \frac{\omega_R}{k}\right) \right] \end{aligned}$$

We want to find the derivative of f_1

$$\begin{aligned} \frac{\partial f_1}{\partial v} &= \frac{\partial f_0}{\partial v} \beta_k \cos(kx - kv t) + f_0 \beta_k k t \sin(kx - kv t) + \frac{eE_0}{m_e} \frac{d^2 f_0}{dv^2} \cdot \\ &\quad \cdot \left[t \sin(kx - \omega_R t) + \frac{1}{2} k t^2 \cos(kx - \omega_R t) \left(v - \frac{\omega_R}{k}\right) \right] + \\ &\quad + \frac{eE_0}{m_e} \frac{df_0}{dv} \left[-\frac{1}{2} k t^2 \cos(kx - \omega_R t) \right] \end{aligned}$$

The terms which goes like t^2 are prevalent for large t ($t \rightarrow \infty$).

However, the term $(v - \omega_R/k)$ is small because we are very close to the pole, so the dominant term for $t \rightarrow \infty$ is the last one.

So

$$\left| \frac{df_0}{dv} \right| \approx \frac{eE_0}{m_e} \frac{df_0}{dv} \frac{1}{2} k t^2 \implies 1 \approx \frac{eE_0}{m_e} \frac{1}{2} k t^2$$

(indeed the cosine oscillates and we can approximate it to 1)

From the last expression we can define the **trapping time**

$$\tau \equiv \sqrt{\frac{m_e}{eE_0 k}}$$

At the trapping time, the linear approximation is not valid anymore.

Energetics of wave-particle resonant interaction

Is the energy conserved? To prove it, we assume that it is conserved.

The energy balance is the following, with K kinetic energy of the particles and W wave energy

$$\frac{dK}{dt} = -\frac{dW}{dt}$$

Kinetic energy density

$$K = \frac{m_e}{2} \int_{-\infty}^{\infty} v^2 f_1(x, v, t) dv$$

recalling

$$\frac{d}{dt}(vf_0) = 0$$

Using the greek letter " κ " to indicate

$$\kappa = \frac{m_e}{2} \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} dx \int_{-\infty}^{\infty} v^2 f_1(x, v, t) dv$$

with $\lambda = 2\pi/k$ being the wavelength of the perturbation.

$$\frac{d}{dt}\kappa = \frac{m_e}{2} \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} dx \int_{-\infty}^{\infty} v^2 \frac{\partial}{\partial t} f_1(x, v, t) dv$$

We use a galileian transformation to move in the Wave Reference Frame

$$v' = v - \frac{\omega_R}{k}, \quad x' = x - \frac{\omega_R}{k}t, \quad t' = t$$

we have

$$\frac{d}{dt}\kappa = \frac{m_e}{2} \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} dx' \int_{-\infty}^{\infty} \left(v' - \frac{\omega_R}{k}\right)^2 \frac{\partial}{\partial t'} f'_1(x', v', t') dv$$

$$\begin{aligned}
f_1'(x', v', t') &= f_0(v') \beta_k \cos \left[k \left(x' + \frac{\omega_R}{k} t' \right) - k \left(v' + \frac{\omega_R}{k} t' \right) t' \right] + \frac{eE_0}{m_e} \frac{\partial f_0(v')}{\partial v'} \cdot \\
&\cdot \frac{1}{\frac{\omega_R}{k} - k \left(v' + \frac{\omega_R}{k} t' \right)} \left\{ \cos \left[k \left(x' + \frac{\omega_R}{k} t' \right) - \omega_R t' \right] + \right. \\
&\left. - \cos \left[k \left(x' + \frac{\omega_R}{k} t' \right) - k \left(v' + \frac{\omega_R}{k} t' \right) t' \right] \right\} = \\
&= f_0(v') \beta_k \cos(kx' - kv't') + \frac{eE_0}{m_e} \frac{\partial f_0(v')}{\partial v'} \left\{ \frac{\cos(kx') - \cos(kx' - kv't')}{-kv'} \right\}
\end{aligned}$$

Now in order to have an easier notation we change the primed variables to unprimed variables. We apply the averages

$$\frac{\partial}{\partial t} \langle f_1(x, v, t) \rangle = \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} \frac{\partial}{\partial t} f(x, v, t) dx$$

In total linear approximation of the Vlasov equation we obtain $0 = 0$, so we need to consider at least the first nonlinear term

$$\begin{aligned}
\frac{\partial}{\partial t} \langle f_1 \rangle + v \left\langle \frac{\partial f_1}{\partial x} \right\rangle - \frac{e}{m_e} \langle E_1 \rangle \frac{\partial f_0}{\partial v} - \frac{e}{m_e} \left\langle E_1 \frac{\partial f_1}{\partial v} \right\rangle &= 0 \\
\Rightarrow \frac{\partial}{\partial t} \langle f_1 \rangle &= \frac{e}{m_e} \frac{\partial}{\partial v} \langle E_1 f_1 \rangle
\end{aligned}$$

So an increase in the kinetic energy is nonlinear effect

$$\begin{aligned}
\frac{\partial}{\partial t} \langle f_1 \rangle &= \frac{e}{m_e} E_0 \left\langle \sin kx \cdot \frac{\partial f_1}{\partial v} \right\rangle \\
\frac{d}{dt} \kappa &= \frac{m_e}{2} \int_{-\infty}^{\infty} \left(v - \frac{\omega_R}{k} \right)^2 \frac{eE_0}{m_e} \left\langle \sin kx \cdot \frac{\partial f_1}{\partial v} \right\rangle dv = \\
&= \frac{m_e}{2} \left[\left(v - \frac{\omega_R}{k} \right)^2 \frac{eE_0}{m_e} \langle \sin kx \cdot f_1 \rangle \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{\infty} 2 \left(v - \frac{\omega_R}{k} \right) \frac{eE_0}{m_e} \langle \sin kx \cdot f_1 \rangle dv \right]
\end{aligned}$$

Considering that f_1 goes to zero for v which goes to infinity

$$f_1(x, v \rightarrow \pm\infty, t) \rightarrow 0$$

and simplifying the factor $m_e/2$, we get

$$\frac{d}{dt} \kappa = -eE_0 \int_{-\infty}^{\infty} \left(v - \frac{\omega_R}{k} \right) \langle \sin kx \cdot f_1 \rangle dv$$

From this expression two integrals come out.

The first integral is

$$\frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} \sin(kx) \cos(kx) dx = \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} \sin(kx) d(\sin(kx)) = \frac{\sin^2(kx)}{2} \cdot \frac{k}{2\pi} \Big|_{-\pi/2}^{\pi/2} = 0$$

The second integral is (recalling that $\lambda = 2\pi/k$)

$$\begin{aligned}
\frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} \sin(kx) \cos(kx - kvt) dx &= \\
&= \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} \sin(kx) \left\{ \cancel{\cos(kx) \cos(kvt)} + \sin(kx) \sin(kvt) \right\} dx = \\
&= \frac{1}{\lambda} \sin(kvt) \int_{-\lambda/2}^{\lambda/2} \sin(kx) \sin(kx) dx = \\
&= \frac{\sin(kvt)}{2}
\end{aligned}$$

So

$$\begin{aligned}
\frac{d}{dt} \kappa &= -eE_0 \int_{-\infty}^{\infty} \left(v - \frac{\omega_R}{k} \right) \langle \sin kx \cdot f_1 \rangle dv = \\
&= -eE_0 \int_{-\infty}^{\infty} \left(v - \frac{\omega_R}{k} \right) \left\{ f_0(v) \beta_k \frac{\sin(kvt)}{2} + \frac{eE_0}{m_e} \frac{\partial f_0}{\partial v} \frac{\sin(kvt)}{2kv} \right\} dv
\end{aligned}$$

Which is the resonant function

We will need the following definition of the Dirac delta

$$\delta(x) = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi x} \sin\left(\frac{x}{\varepsilon}\right)$$

so

$$\lim_{t \rightarrow \infty} \frac{\sin(kvt)}{kv} \approx \pi \delta(kv)$$

We can rewrite the first term as

$$\int_{-\infty}^{\infty} \left(v - \frac{\omega_R}{k} \right) f_0(v) \beta_k \frac{kv}{2} \pi \delta(kv) dv = 0$$

this term is zero because the delta is different from zero for $kv = 0$, but in this point the function which multiplies the delta is null.

Now we recall the following property of the Dirac delta

$$\delta(Ax) = \frac{1}{|A|} \delta(x)$$

The second term is

$$-\frac{e^2 E_0^2}{2m_e} \int_{-\infty}^{\infty} \left(v + \frac{\omega_R}{k} \right) \frac{\partial f_0}{\partial v} \pi \delta(kv) dv = -\frac{e^2 E_0^2}{2m_e} \int_{-\infty}^{\infty} \left(v + \frac{\omega_R}{k} \right) \frac{\partial f_0}{\partial v} \frac{\pi}{k} \delta(v) dv$$

So, in the wave reference frame, we have

$$\frac{d}{dt} \kappa = -\frac{e^2 E_0^2}{2m_e} \frac{\pi}{k} \left(v + \frac{\omega_R}{k} \right) \frac{\partial f_0}{\partial v} \bigg|_{v=0}$$

Resonant function

We see the main contribution close to $v = 0$, which is the resonant region of the wave.

Now we have a look at

$$\frac{d}{dt}W$$

This time we have

$$E_1(x, t) = E_0 \sin(kx) \exp(\omega_I t)$$

The energy density associated to an electric wave (in cgs) is

$$\begin{aligned} W &= \frac{E_1^2}{4\pi} = \\ &= E_0^2 \frac{e^{2\omega_I t}}{4\pi} \frac{1}{\lambda} \int_{-\lambda/2}^{\lambda/2} \sin^2(kx) dx = \\ &= E_0^2 \frac{e^{2\omega_I t}}{4\pi} \frac{k}{2\pi} \frac{1}{k} \int_{-\pi/4}^{\pi/4} \sin^2(kx) d(kx) = \\ &= E_0^2 \frac{e^{2\omega_I t}}{4\pi} \int_{-\pi}^{\pi} \sin^2(y) dy = \\ &= E_0^2 \frac{e^{2\omega_I t}}{4\pi} \frac{1}{2} [y - \sin y \cos y] \Big|_{-\pi}^{\pi} = \\ &= \frac{E_0^2 e^{2\omega_I t}}{8\pi} \end{aligned}$$

and

$$\frac{d}{dt}W = \frac{E_0^2}{8\pi} 2\omega_I e^{2\omega_I t} \simeq \frac{E_0^2}{4\pi} \omega_I$$

so

$$\frac{dK}{dt} = -\frac{dW}{dt} \implies -\frac{e^2 E_0^2}{2m_e} \pi \frac{\omega_R}{k^2} \frac{\partial f_0}{\partial v} \Big|_{v=0} = -\frac{E_0^2}{4\pi} \omega_I$$

from which it follows that

$$\omega_I = \frac{4\pi e^2}{m_e} \frac{\pi}{2} \frac{\omega_R}{k^2} \frac{\partial f_0}{\partial v} \Big|_{v=0} \cdot \frac{n_0}{n_0} = \omega_{pe}^2 \frac{\pi}{2} \frac{\omega_R}{n_0 k^2} \frac{\partial f_0}{\partial v} \Big|_{v=0}$$

and, in the laboratory frame, we have

$$\omega_I^{(\text{LAB})} = \omega_{pe}^2 \frac{\pi}{2} \frac{\omega_R}{n_0 k^2} \frac{\partial f_0}{\partial v} \Big|_{v=v_\phi}$$

which is the Landau result.

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